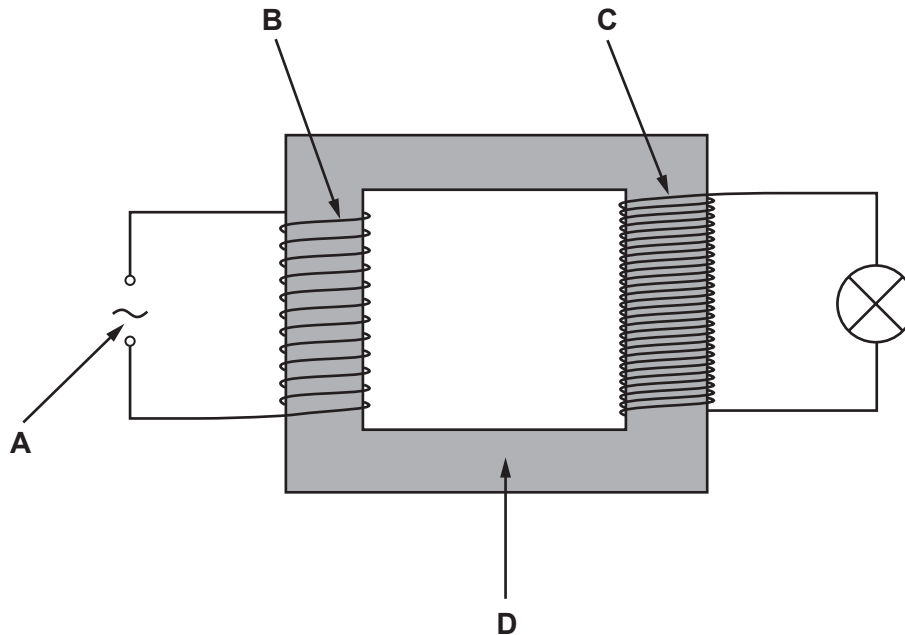


Answer **all** questions.

1. The diagram shows a transformer that is used in a laboratory experiment.



- (a) Draw one line from each letter to the correct label.

[3]

A	primary coil
B	laminated iron core
C	secondary coil
D	a.c. input (supply)

- (b) The diagram shows a step-up transformer. Complete the sentences by underlining the correct word or phrase in the brackets. [3]

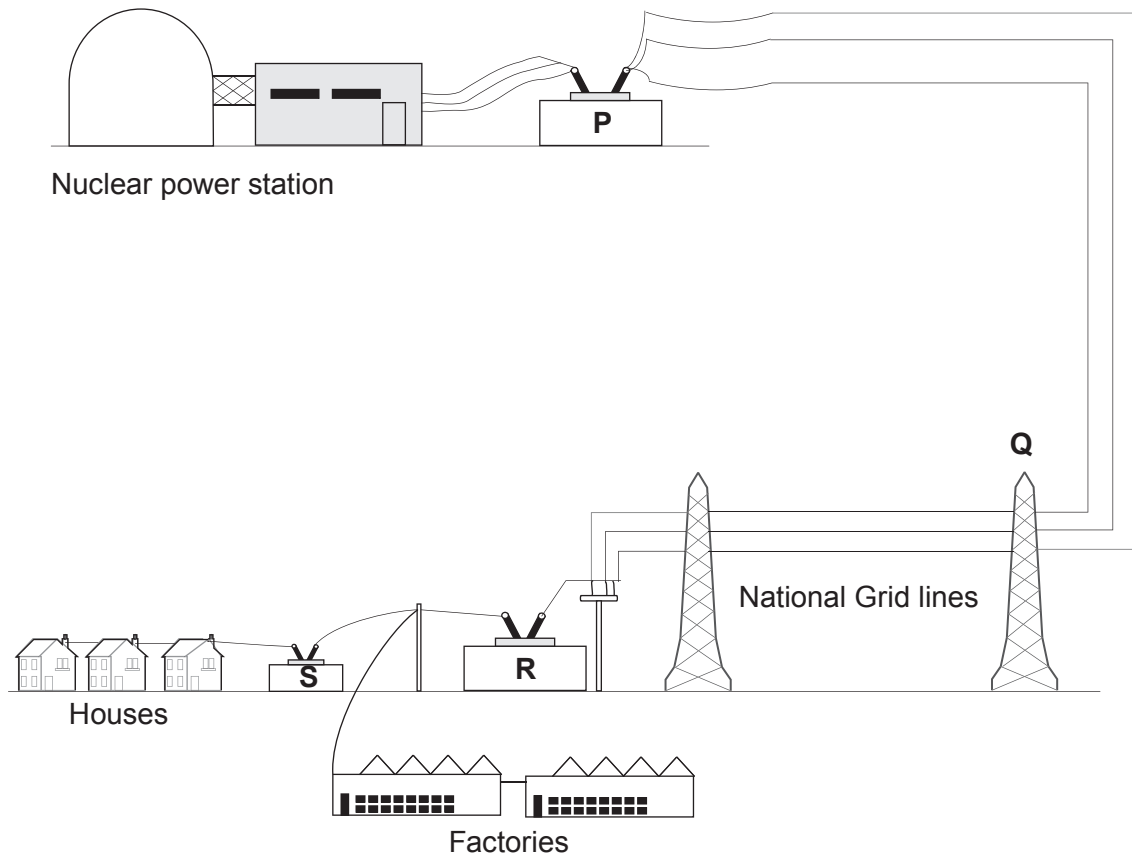
A step-up transformer that is 100% efficient will:

- make the output voltage (**smaller / stay the same / bigger**).
- make the output current (**smaller / stay the same / bigger**).
- make the output power (**smaller / stay the same / bigger**).

- (c) The diagram below shows part of the National Grid. Power stations are used to generate electricity. They are linked to houses and factories by a network of cables.

(i) Which letter **P**, **Q**, **R** or **S** shows a step-up transformer?

[1]



- (ii) State **one** environmental advantage of using a nuclear power station compared to a gas-fired power station. [1]

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